



MSc Business Analytics Dissertation Portfolio MGT7219

Detecting Fraudulent Job Postings Using Machine Learning and Text Mining Techniques

Name: Uzair Abdullah Khan

Student ID: 40385187

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Objective and Purpose

The internet is now flooded with fraudulent job offers, posing a significant hazard to both job searchers and companies due to the surge in the number of such postings. There are some platforms that scammers use to create bogus job offers that look real to the victims within the job market (Mehta et al., 2022). The aim of this study is to utilize machine learning methods to attain a high level of predictive accuracy in differentiating between fraudulent and authentic job adverts.

Primary Question:

- Among the machine learning models, which one exhibits the highest accuracy in categorizing the counterfeit job postings?

Sub-questions:

- It is crucial to identify the primary domains of discussion when differentiating between legitimate job opportunities and fraudulent schemes.
- What text characteristics influence the detection of fraudulent job postings and to what degree?
- How effective are natural language processing techniques in enhancing the detection of fraudulent job offers?

The first step involves reviewing the existing research on detecting job posting fraud. Subsequently, various machine learning techniques, including K-nearest neighbours, will be employed to examine and predict the authenticity of job offers as observed by Huang et al. (2020) which is a well-known parametric technique. The project will utilize the Cross Industry Standard Process for Data Mining (CRISP-DM) as the paradigm. As part of the evaluation of the practicality of data analysis, the earliest steps involve providing fundamental explanations and developing preliminary models. According to Hicks & Peng (2019), statistical model along with abovementioned general framework has been used to determine the success of data analysis as it was suggested to have structure to assess the analysis outcomes. The purpose of this project is to further our understanding of how machine learning may be used to ensure the safety and integrity of the employment market.

Literature Review

The objective of this chapter is to discuss the previous studies and literature on how fake job advertisement can be identified. This section has three subtopics. First, this research focuses on the problem of analysing the prospects for identifying fake vacancies with their effects on both the target audience and the overall job market. Second, it examines the nature of fake employment ads; various aspects are studied, including the general properties of the fake job offers and the patterns typical of fake ads. Finally, this chapter discusses machine learning for fraud detection, presenting prior studies' approaches and performance measures.

2.1 The Importance of Identifying Counterfeit Job Advertisements

It is imperative to have strong detection techniques due to the substantial dangers posed by the rising number of fake job ads on the internet. Carmen et al (2010) said that failure to search for appropriately spelled words, ambiguous information and wrong details in jobs advertised is a failure to look for signs of suspect activity. Counterfeit employment advertisements can result in the theft of one's identity, financial detriment, and the compromise of personal data. Prior research emphasizes the significance of employing sophisticated technologies to alleviate these dangers. Job posting fraud can be influenced by factors such as linguistic trends, job specifications, and posting frequency. As per M.S.R et al. (2024) these fraudulent postings have been on the rise due to the proliferation of online job platforms. Identifying deceptive job listings is crucial for upholding the credibility of online job platforms and safeguarding job seekers.

2.2 Features of Counterfeit Employment Advertisement

Studies suggest that counterfeit job advertisements frequently have noticeable traits in contrast to authentic ones. The approaches to identify fake jobs include verifying the demands for individual details, identifying deficiencies in the ad wording, raising awareness on jobs with overemphasized rewards, and tracking incessant communications on online media (Hanif, 2024). These encompass ambiguous job descriptions, impractical wage propositions, and dubious contact details. Machine learning algorithms can utilize these characteristics to differentiate between authentic and deceptive job advertisements. Text mining and natural language processing (NLP) are crucial in discovering these characteristics by analysing the syntax and semantics of job advertisements.

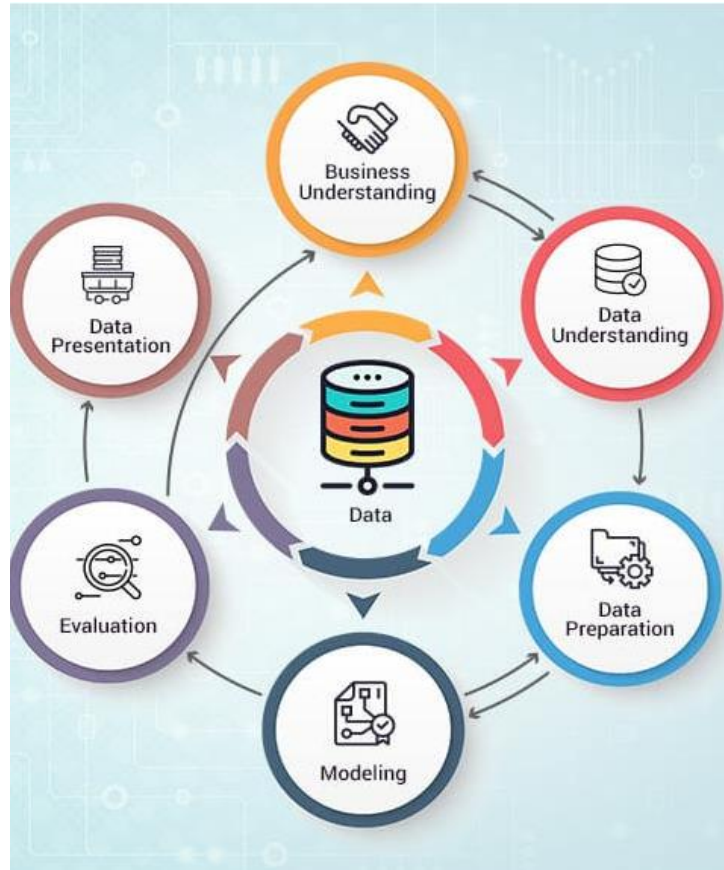
2.3 Utilizing Machine Learning Techniques for the Purpose of Fraud Detection

Diverse machine learning techniques have been utilized to identify fraudulent job advertisements. Methods such as logistic regression, decision trees, support vector machines (SVM), and K-nearest neighbours (KNN) have demonstrated encouraging outcomes. Moreover, it has also been proved that the combination of decision trees and random forest algorithm can be useful in accurately identifying the fraudulent job postings on the web (Srivastava, 2022) Also, these systems can utilize single classifiers and ensemble classifiers in order to evaluate possible results in the attempt to differentiate between real and fake job offers (Ullah & Jamjoom, 2023; Alandjani, 2022; Dutta & Bandyopadhyay, 2020). Research has shown that the integration of natural language processing (NLP) with machine learning significantly improves the accuracy of detection (Kabir et al. 2022). Logistic regression and SVM are proficient in binary classification tasks (Ghavamipour et al., 2021), however KNN offers a versatile method for dealing with intricate patterns in the data.

Ultimately, this literature review seeks to consolidate significant discoveries about the identification of fraudulent job advertisements through the utilization of machine learning. This study aims to discover the most effective options for enhancing job market security by analysing and evaluating existing strategies and their effectiveness.

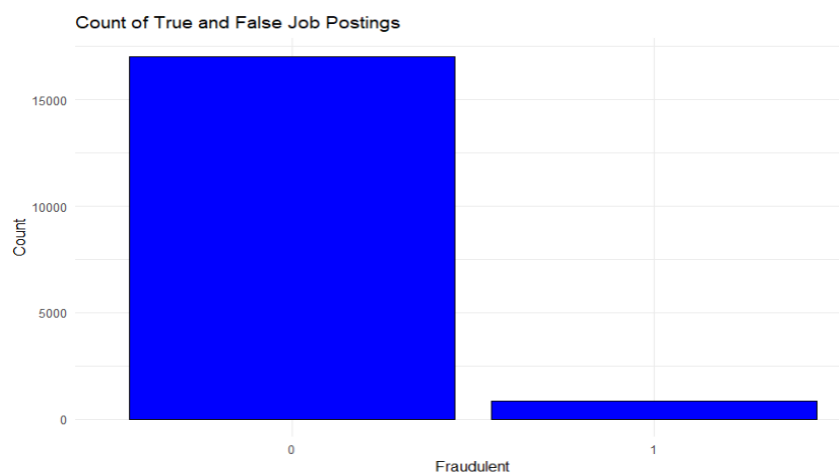
Methodology:

The technique used in this proposal adheres to the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework. CRISP-DM offers a versatile and methodical framework that is well-suited for data mining initiatives in diverse industries.



3.1 Data Gathering and Understanding

The dataset applied in this study was uploaded to Kaggle in year 2020. Employment Scam Aegean is a website that was originally used to collect the records and it includes nearly 10000 real life job posting out of which only around 800 are fake. It includes job related textual data as well as meta-data about those jobs. The dataset is imbalanced; the ratio of real and fake job offers on the dataset is approximately 20:1 as shown in the image below.



Following table describes the columns used in the data.

Column Name	Data Type	Missing Values	Data
job_id	int	0	Unique Job IDs
title	char	0	Job Title
location	char	346	Location (City, State)
department	char	11547	Department
salary_range	char	15012	Expected Salary Ranges
company_profile	char	3308	Company Description
description	char	0	Job Description
requirement	char	2694	Job Requirements
benefits	char	7208	Company perks and benefits
telecommuting	int	0	Home based or Office based
has_company_logo	int	0	1 for yes, 0 for no
has_questions	int	0	1 for yes, 0 for no
employment_type	char	3471	Part-time, Full-time, contract, permanent etc
required_experience	char	7050	Job experience level required (mid-level/associate etc)
required_education	char	8105	Minimum education required (vocational/high school, etc)
industry	char	4903	Job industry (Banking/marketing, etc)
function	char	6455	Job function
fraudulent	int	0	1 for fraud, 0 for real

3.2 Data Preprocessing

To tackle the problem of identifying and handling missing values in character columns within a dataset, we first applied a function to each column to count the missing values. Next, we transformed this count data into a data frame, making it suitable for visualization. This plot provided a clear and immediate visual summary of the missing data landscape in our dataset.

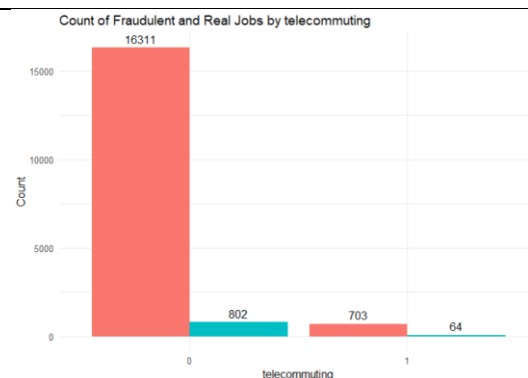
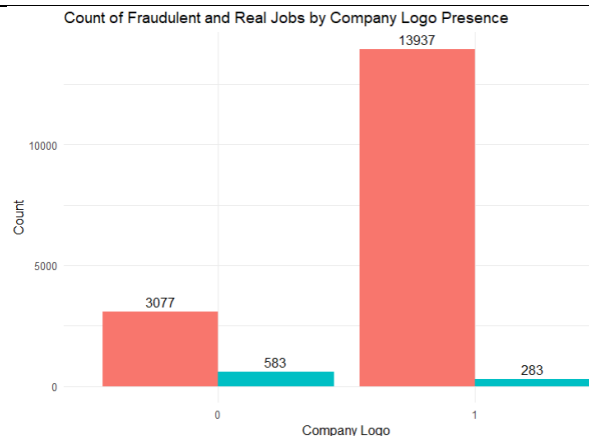
To further enhance the dataset's usability, a preprocessing function was developed to clean and normalize textual data. This function converted text to lowercase, stripped HTML tags, removed punctuation, tokenized the text, filtered out stopwords, and applied stemming. Additionally, it ensured that the text length did not exceed Excel's character limit. This function was then applied to multiple columns in the dataset using mutate, resulting in a cleaned version of the data.

Finally, we converted several key columns to factors, ensuring they were treated as categorical data for subsequent analysis. The structure of the dataset was verified using the str function to confirm that the transformations were successful, and that the data was ready for further modelling or analysis. This comprehensive preprocessing pipeline set the stage for robust data analysis and machine learning tasks.

3.3 Data Visualizations

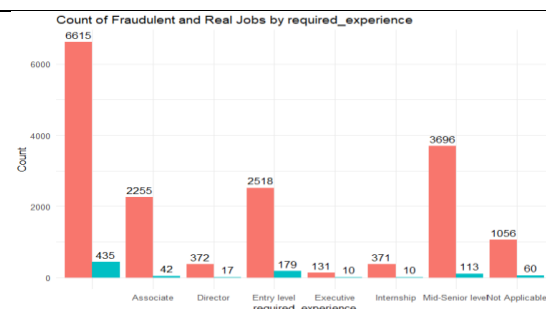
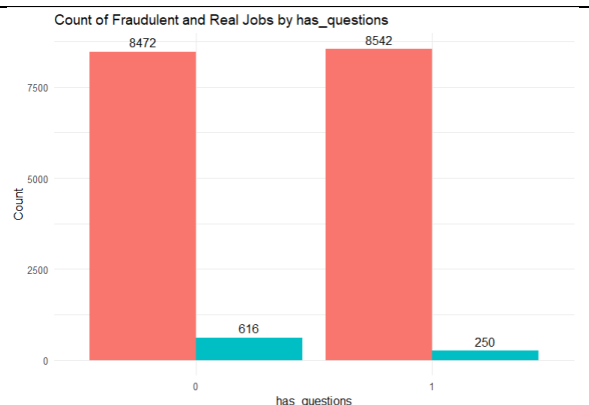
Data visualizations were utilized to acquire an initial comprehension of the dataset prior to implementing any statistical approaches or modelling. They were performed on the columns that had the values as factors to see if any patterns or potential relationships could be observed in these. The variables selected were has_company_logo, has_questions, telecommuting, required_experience, required_education and employment_type. As it can be observed in the shared images there were some patterns that could be observed.

As it can be observed, the presence of company logo reduces the likelihood of the job being fraudulent significantly.



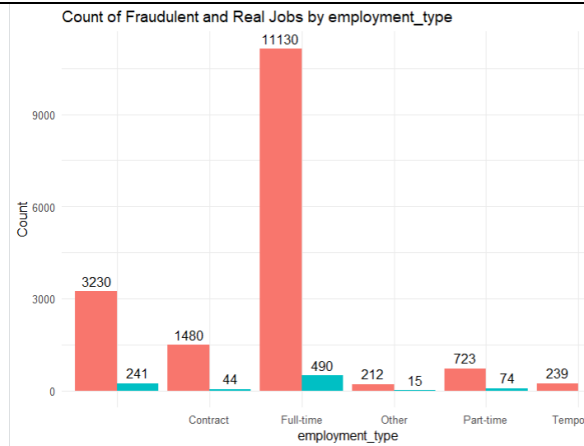
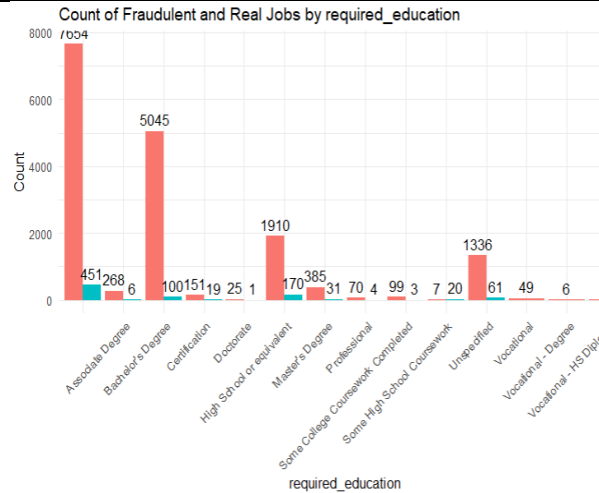
Similarly, the jobs being advertised as home based are up to twice more likely to be fraudulent as compared to the ones advertised office based.

Likewise, it can be observed that the jobs that have questions are 3 times more likely to be real ones.



Based on the required_experience, jobs being advertised as entry level are most likely to be fake compared to other required experience levels. Moreover, jobs for mid-senior level experience are most likely to be legitimate.

In a similar fashion, jobs requiring a minimum bachelor's level education are rarely fake. As per the given data even less than 2% of such jobs were fraud compared to 13% of jobs which only required a certification.



Contract Jobs were found to be more legit along with full-time jobs compared to part time jobs which were 3 times more likely to be fraud.

3.4 Data Modelling

Numerous machine learning algorithms can be implemented to flag the fraudulent jobs being posted. However, this study will employ the most effective techniques for the purpose of classifying and identifying fraudulent job postings.

1. Logistic regression

A linear model used for binary classification that estimates the likelihood of a binary result by considering input features (Manjunatha et al., 2022).

Logistic regression is ideal for binary classification and can handle large datasets with 17,000 job postings. It provides a straightforward way to analyse the relationship between text features (job description, company profile, job requirements, job benefits) and the likelihood of a job posting being fraudulent.

2. Support Vector Machines(SVMs)

This classifier identifies the most effective hyperplane that maximizes the distance between different classes in a space with several dimensions (Coolen, 2005).

SVMs are effective in high-dimensional spaces, making them suitable for text data with many features. The ability to use kernel tricks helps SVMs manage non-linear relationships in the selected dataset, enhancing accuracy in distinguishing between real and fake job postings.

3. Random Forests

It is a collection of decision trees, with each tree being trained on a bootstrap sample of the data (Breiman, 2001, Strobl et al., 2008).

Random Forests handle large datasets and missing values efficiently. They provide feature importance metrics, which can help identify key indicators of fraudulent job postings. Their ensemble nature improves robustness and accuracy.

4. Gradient Boosting algorithms

This is an ensemble technique that constructs trees in a sequential manner, with each tree aimed at rectifying the flaws made by the preceding trees (Natekin & Knoll, 2013). Moreover, XG boost, being considered an optimized gradient boosting algorithm, is pronounced as more efficient and porous to different kinds of datasets than most other algorithms (Konda et al. , 2021).

Gradient Boosting algorithms excel at handling imbalanced datasets, such as given dataset with 800 fake jobs out of 17,000. They iteratively correct errors from previous models, capturing complex patterns in text data for better prediction accuracy.

5. Naive Bayes

Naive Bayes is one of the most common classifiers that is characterized by simplicity and effectiveness in many tasks. This is assumed since attributes in a certain class are independent of each other if the class label is provided (Xie et al. , 2002). Thus, it has been used in document classification, e-mail spam filtering, sentiment analysis and other classification (Wicaksono, 2023; Rissan & Hassan, 2022)

It is highly effective for text classification tasks due to its simplicity and efficiency with large vocabularies. It leverages word frequencies, making it suitable for identifying patterns in job descriptions and other text fields.

6. Artificial Neural Networks (ANN)

A versatile model consisting of interconnected neurons that may effectively capture intricate non-linear interactions(Deulofeu et al., 2019).

ANNs are suitable for handling large and complex datasets with intricate non-linear patterns, which is typical in text data. They can learn and adapt to the complexities in job descriptions, company profiles, job requirements, and job benefits, making them effective for detecting fraudulent job postings. With enough training data and computational resources, ANNs can achieve high accuracy. This algorithm is being kept as a back-up option to see if it can improve the efficiency and make the model more accurate.

The study seeks to utilize machine learning approaches to create strong models that can effectively identify fraudulent job listings. These models will leverage the unique capabilities of each method to handle different parts of the data.

There are some other machine learning techniques available as well which were explored, however, it was decided not to move forward with them because of the mentioned reasons.

1. K-Nearest Neighbours (KNN)

This is a non-parametric classification method that determines the class of a data point based on the majority class of its nearest neighbors.

KNN is computationally expensive for large datasets and high-dimensional text data. It struggles with scalability and efficiency when processing 17,000 job postings.

2. Decision Trees

A hierarchical arrangement in which the internal nodes represent feature tests, and the leaves reflect outcomes. Decision Trees tend to overfit and may not capture complex relationships in text data as effectively as ensemble methods like Random Forest or Gradient Boosting (Stahl & Bramer, 2012) .

3. Polynomial regression

This regression model enhances linear regression by using polynomial terms of the predictors.

Polynomial Regression is more suited for continuous data and non-linear relationships, rather than binary classification tasks involving text data.

4. Lasso regression

This is a linear regression model that incorporates L1 regularization. L1 regularization encourages sparsity in the model by reducing less important coefficients to zero. While Lasso helps with feature selection and regularization, it is less effective than ensemble methods for capturing complex interactions in text data, which are crucial for detecting fraudulent jobs.

5. Ridge regression

This is a linear regression model that incorporates L2 regularization. L2 regularization is a technique that penalizes the size of the coefficients to mitigate multicollinearity.

Like Lasso, Ridge Regression is better suited for addressing multicollinearity in continuous data rather than binary classification tasks with textual features (Merkwirth et al., 2004). Ensemble methods like Random Forest and Gradient Boosting provide more robustness and accuracy for your specific use case.

Conclusion

In conclusion, the analysis conducted in this project shows that the data is appropriate for meeting the project's goals. By doing thorough data investigation and analysis, we have obtained significant insights. The initial results suggest that the use of text columns such as job descriptions, company profiles, job requirements, and job benefits greatly improves the identification of bogus job posts. Utilizing machine learning methods such as Logistic Regression, SVMs, Random Forests, Gradient Boosting, and Artificial Neural Networks has established a strong basis for creating reliable models that enhance job market stability and decrease the occurrence of fraudulent job listings.

Project Plan

After the submission of the research proposal, I shall conduct the literature review on the research topic in the first and second week of July. At the same time, this study will help in selecting and comparing several machine learning algorithms with the intention of increasing the effectiveness of the models. The feedback I am going to get from the supervisor will comprise criticisms and directions that will help improve the quality of research. The next step is to conduct empirical studies and conduct data analysis for the results obtained. In July the models of machine learning shall be constructed to solve the research inquiries in week three and week four of the course. The first drafts of the research report, technical report, and logbook will be complete by the end of July. The final report will then be produced in mid – August after the complete review and proof – reading. However, the above initiative has the following risks that are inherent in the organizational environment. One potential risk of this research study is that of sample data quality. This is because, when there is missing data or data which does not tally with the researcher’s study, the whole research process may be affected and delayed. Also, since the research assumes the use of complex data analysis methods and machine learning algorithms, there may be certain technical challenges. We can find out these hindrances as a way of delaying the progress of the project.

July	Week 1	Submit Proposal and Reading Previous Literature
	Week 2	Meet with Supervisor and Select Test methods
	Week 3	Working with models and data analysis
	Week 4	Getting results from initial tests to see if the model accuracy can be improved
August	Week 1	Testing further models to check the accuracy being improved.
	Week 2	Drafting the research report and technical report with discussions
	Week 3	Feedback from Supervisor before final edits
	Week 4	Final Edits and Proof Reading
September	Week 1	Submission

Appendix A: References

- Alandjani, G. (2022). Online fake job advertisement recognition and classification using machine learning. 3C TIC: Cuadernos De Desarrollo Aplicados a Las TIC, 11(1), 251-267. <https://doi.org/10.17993/3ctic.2022.111.251-267>
- Breiman, L. (2001). Untitled. Machine Learning, 45(1), 5-32. <https://doi.org/10.1023/a:1010933404324>
- Coolen, A. (2005). Support vector machines for binary classification., 185-204. <https://doi.org/10.1093/oso/9780198530237.003.0008>
- Deulofeu, M., Kolářová, L., Salvadó, V., Peña-Méndez, E., Alması, M., Štork, M., ... & Vaňhara, P. (2019). Rapid discrimination of multiple myeloma patients by artificial neural networks coupled with mass spectrometry of peripheral blood plasma. Scientific Reports, 9(1). <https://doi.org/10.1038/s41598-019-44215-1>
- Dutta, S. and Bandyopadhyay, S. K. (2020). Fake job recruitment detection using machine learning approach. International Journal of Engineering Trends and Technology, 68(4), 48-53. <https://doi.org/10.14445/22315381/ijett-v68i4p209s>
- Ghavamipour, A. R., Türkmen, F., & Jian, X. (2021). Privacy-preserving logistic regression with secret sharing.. <https://doi.org/10.48550/arxiv.2105.06869>
- Hanif, A. H., Maarop, N., Kamaruddin, N., & Samy, G. N. (2024). Machine learning approach in predicting fraudulent job advertisement. International Journal of Academic Research in Business and Social Sciences, 14(1). <https://doi.org/10.6007/ijarbss/v14-i1/20532>
- Hicks, S. C. and Peng, R. D. (2019). Evaluating the success of a data analysis.. <https://doi.org/10.48550/arxiv.1904.11907>
- Huang, S., Huang, M., & Lyu, Y. (2020). An improved knn-based slope stability prediction model. Advances in Civil Engineering, 2020, 1-16. <https://doi.org/10.1155/2020/8894109>
- Kabir, M. K., Islam, M., Kabir, A. N. B., Haque, A., & Rhaman, M. K. (2022). Detection of depression severity using bengali social media posts on mental health: study using natural language processing techniques. JMIR Formative Research, 6(9), e36118. <https://doi.org/10.2196/36118>
- Konda, N., Verma, R., & Jayaganthan, R. (2021). Machine learning based predictions of fatigue crack growth rate of additively manufactured ti6al4v. Metals, 12(1), 50. <https://doi.org/10.3390/met12010050>
- Manjunatha, K., Agarwal, V., Mack, A., Koester, D., & Adams, D. (2022). Total unwrapped phase-based diagnosis of wall thinning in nuclear power plants secondary piping structures. IEEE Access, 10, 113726-113740. <https://doi.org/10.1109/access.2022.3212061>
- Mehta, K., Rajagopal, N. K., Gaikwad, S. B., & Yadav, S. (2022). Machine learning techniques for true and fake job posting. ECS Transactions, 107(1), 2383-2389. <https://doi.org/10.1149/10701.2383ecst>
- Merkwirth, C., Mauser, H., Schulz-Gasch, T., Roche, O., Stahl, M., & Lengauer, T. (2004). Ensemble methods for classification in cheminformatics. Journal of Chemical Information and Computer Sciences, 44(6), 1971-1978. <https://doi.org/10.1021/ci049850e>

- M. S. R., -, M. H. L., -, L. S., & -, S. A. (2024). Fake job post detection using machine learning. *International Journal for Multidisciplinary Research*, 6(1). <https://doi.org/10.36948/ijfmr.2024.v06i01.13906>
- Natekin, A. and Knoll, A. (2013). Gradient boosting machines, a tutorial. *Frontiers in Neurorobotics*, 7. <https://doi.org/10.3389/fnbot.2013.00021>
- Rissan, M. and Hassan, R. (2022). Naïve-bayes family for sentiment analysis during covid-19 pandemic and classification tweets. *Indonesian Journal of Electrical Engineering and Computer Science*, 28(1), 375. <https://doi.org/10.11591/ijeecs.v28.i1.pp375-383>
- Srivastava, R. (2022). Identification of online recruitment fraud (orf) through predictive models. *Emirati Journal of Business, Economics, & Social Studies*, 1(1), 42-54. <https://doi.org/10.54878/epv1hr57>
- Stahl, F. and Bramer, M. (2012). Jmax-pruning: a facility for the information theoretic pruning of modular classification rules. *Knowledge-Based Systems*, 29, 12-19. <https://doi.org/10.1016/j.knosys.2011.06.016>
- Strobl, C., Boulesteix, A., Kneib, T., Augustin, T., & Zeileis, A. (2008). Conditional variable importance for random forests. *BMC Bioinformatics*, 9(1). <https://doi.org/10.1186/1471-2105-9-307>
- Ullah, Z. and Jamjoom, M. (2023). A smart secured framework for detecting and averting online recruitment fraud using ensemble machine learning techniques. *PeerJ Computer Science*, 9, e1234. <https://doi.org/10.7717/peerj-cs.1234>
- Wicaksono, A. (2023). Twitter sentiment analysis of electric vehicle subsidy policy using naïve bayes algorithm. *JSMDs*, 1(1), 1-9. <https://doi.org/10.31258/jsmds.v1i1.3>
- Xie, Z., Hsu, W., Liu, Z., & Lee, M. (2002). Snnb: a selective neighborhood based naïve bayes for lazy learning., 104-114. https://doi.org/10.1007/3-540-47887-6_10

Appendix B: Completed ethical approval form



**QUEEN'S
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**QUEEN'S
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SCHOOL**

Student Ethical Approval Form

Name of student: Uzair Abdullah Khan

Student e-mail: ukhan03@qub.ac.uk

Dissertation title: Detecting Fraudulent Job Postings Using Machine Learning and Text Mining Techniques

Section One – Overview of the Research Methodology. In this section you are required to provide an overview of your proposed research methodology. Areas which should be discussed include:

Data Sources:

- ✓ A description of the data set (s) that will be used in the project (what is included in the data, does it relate to identifiable individuals / organisations, how have you ensured the authenticity and quality of the data).
- ✓ Information about the source of the data (whether it is publicly available, who collected it, are there any usage restrictions).
- ✓ Are there any special ethical issues with the proposed data (e.g. does it include children, vulnerable adults, or people with special communication needs)

Method

- ✓ A description of how the research will be carried out and the procedures that you are intending to use (e.g. data visualisation, predictive analytics, text mining).

The data is publicly available at Kaggle since 2020. It was published by Employment Scam Aegean which collected the data. There are no usage restrictions for this data.

The data does not contain any personal information of any individual and even does not contain any column of the company name. The quality of data, although a bit skewed, is very good. There are several missing entries, however, they could be ignored while using machine learning techniques.

The Research will be carried out using several machine learning techniques to find the most accurate way to predict if the posted job is fraud or not. The Crisp -DM methodology would be used throughout the project to keep it compliant with the best practices.

Section 2: Checklist of ethical issues

For each of the questions below, please select Yes or No, as appropriate.

1. Please confirm that the project uses secondary data only (any deviation from this requires prior approval from the programme director along with additional ethics approval paperwork):

Yes

2. Does the study require access to data which is not publicly available?

No

3. Will the study require the co-operation of a gatekeeper for initial access to the data? (E.g. businesses or other organisations)

No

4. Does the project involve data in which individuals or organisations can be identified?

No

5. Does the study involve data relating to participants who are particularly vulnerable or unable to give informed consent? (e.g. Children, people with learning disabilities, or staff/student records)

No

6. Will it be necessary for participants to take part in the study without their knowledge and consent at the time? (E.g. covert observation of people in non-public places)

No

7. Will the study involve data relating to sensitive topics (e.g. sexual activity, drug use)?

No

If you have answered yes to any of the questions, or if there are other ethical issues or considerations, please explain these below:

The data used was collected from secondary source and is publicly available.

Section 3: Management of routine ethical issues

Describe the ways in which the research design has addressed any ethical issues identified in section two (Refer to data security and privacy, informed consent, handling of sensitive data/information).

If there are no ethical issues, provide a full explanation as to why this is the case (e.g. non-identifiable secondary data only is being used and does not include sensitive information).

No ethical issues are present in this research since this work only uses secondary non-identifiable data that will not contribute to the exposure of the respondents' details. The study does not have any volunteers accessing it; thus, no harm, privacy infringement, or violation of confidentiality can happen. The data set applied has been fully anonymized, and therefore did not contain any elements that could lead to the reidentification of any company or organization. Furthermore, the research follows all the current ethic principles and legal precedent, erasing any possible abuse of people's rights and keeping a high standard of ethical research. Therefore, the research project is ethical and is free from any ethical dilemmas.

How has your research design taken into account issues of risk (for example the possibility of being unable to gain access to data or data loss or disclosure of sensitive information)?

To effectively handle potential hazards, the following research design aspects are suggested protections. Firstly, this study has established a standardized system for accessing. Moreover, this research employs a comprehensive approach by incorporating a substantial amount of data. Ultimately, this research has employed appropriate data management techniques, including the removal of identifiable information. Ensuring the preservation of safe storage.

Section Four: Student declaration

I have read and agree to abide by the requirements of the following University documents

- Policy and Principles on the Ethical Approval of Research
<http://www.qub.ac.uk/home/media/Media,599765,en.pdf>
- Code of Conduct and Integrity in Research
<http://www.qub.ac.uk/home/media/Media,599772,en.pdf>

Student's signature: *Uzair Abdullah Khan* Date 05/07/2024

NB: If, as your study progresses, you are proposing to make changes to the data used or other ethical concerns arise, you must bring these to the attention of your supervisor.

Students will be informed at the proposal stage if there are any ethical concerns which should be addressed.

Appendix C: R Code

```
# Install necessary packages

install.packages("tm")

install.packages("textclean")

install.packages("tokenizers")

install.packages("SnowballC")

install.packages("reshape2")


# Load necessary libraries

library(tm)

library(textclean)

library(tokenizers)

library(SnowballC)

library(dplyr)

library(writexl)

library(readxl)

library(ggplot2)

library(dplyr)

library(reshape2)


data <- read.csv("C:/Users/Uzair Khan/Downloads/Assignments/Dissertation/DataSet.csv")


summary(data)


data$fraudulent<- factor(data$fraudulent)


# Create a bar plot for the 'fraudulent' column

ggplot(data, aes(x = fraudulent)) +

  geom_bar(fill = "blue", color = "black") +

  labs(title = "Count of True and False Job Postings",

        x = "Fraudulent",

        y = "Count") +

  theme_minimal()
```

```

# Calculate missing values in character columns
missing_values_char <- sapply(data, function(col) {
  if (is.character(col)) {
    sum(is.na(col) | col == "")
  } else {
    NA
  }
})

# Remove NA values from the result (for non-character columns)
missing_values_char <- missing_values_char[!is.na(missing_values_char)]

print(missing_values_char)

# Convert to data frame for plotting
missing_values_df <- data.frame(
  column = names(missing_values_char),
  missing_count = missing_values_char
)

# Plot the missing values
ggplot(missing_values_df, aes(x = column, y = missing_count)) +
  geom_bar(stat = "identity", fill = "skyblue") +
  geom_text(aes(label = missing_count), vjust = -0.5) +
  labs(title = "Missing Values in Character Columns",
       x = "Column",
       y = "Number of Missing Values") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

summary(data)

# Preprocessing function

```

```

preprocess_text <- function(text) {
  text <- tolower(text)
  text <- replace_html(text)
  text <- removePunctuation(text)
  tokens <- unlist(tokenize_words(text))
  stop_words <- stopwords("en")
  tokens <- tokens[!tokens %in% stop_words]
  tokens <- wordStem(tokens, language = "en")
  processed_text <- paste(tokens, collapse = " ")
  # Truncate if exceeds Excel's limit
  if (nchar(processed_text) > 32767) {
    processed_text <- substr(processed_text, 1, 32767)
  }
  return(processed_text)
}

# Apply preprocessing to multiple columns
clean_data <- data %>%
  mutate(
    company_profile = sapply(company_profile, preprocess_text),
    description = sapply(description, preprocess_text),
    requirements = sapply(requirements, preprocess_text),
    benefits = sapply(benefits, preprocess_text)
  )

# Convert specified columns to factors
clean_data$has_company_logo <- factor(clean_data$has_company_logo)
clean_data$has_questions <- factor(clean_data$has_questions)
clean_data$telecommuting <- factor(clean_data$telecommuting)
clean_data$employment_type <- factor(clean_data$employment_type)
clean_data$required_experience <- factor(clean_data$required_experience)
clean_data$required_education <- factor(clean_data$required_education)
clean_data$fraudulent <- factor(clean_data$fraudulent)

```

```
# Verify the changes
```

```
str(data)
```

```
# Create the plot for has_company_logo
```

```
ggplot(clean_data, aes(x = has_company_logo, fill = fraudulent)) +  
  geom_bar(position = "dodge") +  
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, position = position_dodge(0.9)) +  
  labs(title = "Count of Fraudulent and Real Jobs by Company Logo Presence",  
        x = "Company Logo",  
        y = "Count",  
        fill = "Job Type") +  
  theme_minimal()
```

```
# Create the plot for Telecommuting
```

```
ggplot(clean_data, aes(x = telecommuting, fill = fraudulent)) +  
  geom_bar(position = "dodge") +  
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, position = position_dodge(0.9)) +  
  labs(title = "Count of Fraudulent and Real Jobs by telecommuting",  
        x = "telecommuting",  
        y = "Count",  
        fill = "Job Type") +  
  theme_minimal()
```

```
# Create the plot for has_questions
```

```
ggplot(clean_data, aes(x = has_questions, fill = fraudulent)) +  
  geom_bar(position = "dodge") +  
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, position = position_dodge(0.9)) +  
  labs(title = "Count of Fraudulent and Real Jobs by has_questions",  
        x = "has_questions",  
        y = "Count",  
        fill = "Job Type") +  
  theme_minimal()
```

```
# Create the plot for required_experience
```

```

ggplot(clean_data, aes(x = required_experience, fill = fraudulent)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, position = position_dodge(0.9)) +
  labs(title = "Count of Fraudulent and Real Jobs by required_experience",
       x = "required_experience",
       y = "Count",
       fill = "Job Type") +
  theme_minimal()

```

Create the plot for required_education

```

ggplot(clean_data, aes(x = required_education, fill = fraudulent)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, position = position_dodge(0.9)) +
  labs(title = "Count of Fraudulent and Real Jobs by required_education",
       x = "required_education",
       y = "Count",
       fill = "Job Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

Create the plot for employment_type

```

ggplot(clean_data, aes(x = employment_type, fill = fraudulent)) +
  geom_bar(position = "dodge") +
  geom_text(stat = "count", aes(label = ..count..), vjust = -0.5, position = position_dodge(0.9)) +
  labs(title = "Count of Fraudulent and Real Jobs by employment_type",
       x = "employment_type",
       y = "Count",
       fill = "Job Type") +
  theme_minimal()

```